「Robbins-Siegmund Lemma.

Let $\{V_k\}, \{S_k\}, \{u_k\}$ be non-negative real sequences satisfying

$$V_{k+1} \leq V_k - S_k + u_k \text{ for all } k \in \mathbb{N} \quad \text{and} \quad \sum_{k=0}^{\infty} u_k < \infty$$

Then, $S_k \to 0$ and $\{V_k\}$ has a limit as $k \to \infty$.

Proof. observe that

$$V_{k+1} - V_k + S_k \leq u_k \text{ for all } k \in \mathbb{N}.$$

Therefore, given $N > 1$,

$$\sum_{k=1}^{N} (V_{k+1} - V_k + S_k) = V_{N+1} - V_1 + \sum_{k=1}^{N} S_k \leq \sum_{k=1}^{N} u_k \leq \sum_{k=1}^{\infty} u_k$$

thus $$\sum_{k=1}^{N} S_k \leq \sum_{k=1}^{\infty} u_k + V_1 - V_{N+1} \leq \sum_{k=1}^{\infty} u_k + V_1 < \infty,$$

So $\sum_{k=1}^{\infty} S_k$ converges, $S_k \to 0$ as $k \to \infty$.」

Meanwhile, define

$$W_k = V_k - \sum_{i=1}^{k-1} u_i \quad (k \geq 2).$$

Then, for any $k \geq 2$,

$$W_{k+1} - W_k = V_{k+1} - V_k - u_k \leq -S_k \leq 0,$$ so $\{W_k\}$ is monotone decreasing.

Moreover, given $k \geq 2$,

$$W_k = V_k - \sum_{i=1}^{k-1} u_i \geq -\sum_{i=1}^{k-1} u_i \geq -\sum_{i=1}^{\infty} u_i$$

thus, W_k is bounded below, so converges. Now,

$$V_k = \sum_{i=1}^{k-1} u_i + W_k$$

Converges, being sum of convergent sequences.

「Discrete Robbins - Siegmund Lemma.

Let $\{V_k\}$, $\{a_k\}$, $\{b_k\}$, $\{c_k\}$ be non-negative Real numbers sequences satisfying

$$V_{k+1} \le (1+a_k)\cdot V_k + b_k - c_k \quad \text{and} \quad \sum a_n, \sum b_n < \infty$$

Then, $\sum c_n < \infty$ and $\{V_k\}$ converges.

Proof. observe that

$$c_k \le (1+a_k)\cdot V_k - V_{k+1} + b_k \le (1+a_k)\cdot V_k + b_k$$

[Difference of adjunctions term.

Let $\{z_k\}$ be a series of real numbers.

If $\sum_{n=1}^{\infty} |z_{n+1} - z_n|$ converges, then $\{z_n\}$ converges.

Proof. Given $\varepsilon > 0$, there exist $N \in \mathbb{N}$ s.t.

$$m > n \geq N \Rightarrow \sum_{k=n}^{m} |z_{k+1} - z_k| < \varepsilon.$$

And, by the triangle inequality,

$$|z_m - z_n| \leq |z_m - z_{m-1}| + |z_{m-1} - z_{m-2}| + \cdots + |z_{n+1} - z_n| < \varepsilon.$$

So, $\{z_k\}$ is a Cauchy sequence, so converges.